gular cohomology of K with another kind of cohomology called Čech cohomology. This is defined in the following way. To each open cover $\mathcal{U} = \{U_\alpha\}$ of a given space X we can associate a simplicial complex $N(\mathcal{U})$ called the **nerve** of $\mathcal{U}$. This has a vertex v_α for each U_α, and a set of $k+1$ vertices spans a k-simplex whenever the $k+1$ corresponding U_α's have nonempty intersection. When another cover $\mathcal{V} = \{V_\beta\}$ is a refinement of $\mathcal{U}$, so each V_β is contained in some U_α, then these inclusions induce a simplicial map $N(\mathcal{V}) \to N(\mathcal{U})$ that is well-defined up to homotopy. We can then form the direct limit $\varinjlim H^i(N(\mathcal{U});G)$ with respect to finer and finer open covers $\mathcal{U}$. This limit group is by definition the **Čech cohomology group** $\check{H}^i(X;G)$. For a full exposition of this cohomology theory see [Eilenberg & Steenrod 1952]. With an analogous definition of relative groups, Čech cohomology turns out to satisfy the same axioms as singular cohomology, and indeed a stronger form of excision: a map $(X,A) \to (Y,B)$ that restricts to a homeomorphism $X - A \to Y - B$ induces isomorphisms on Čech cohomology groups. For spaces homotopy equivalent to CW complexes, Čech cohomology coincides with singular cohomology, but for spaces with local complexities it often behaves more reasonably. For example, if X is the subspace of $\mathbb{R}^3$ consisting of the spheres of radius $1/n$ and center $(1/n, 0, 0)$ for $n = 1, 2, \cdots$, then contrary to what one might expect, $H^3(X;\mathbb{Z})$ is nonzero, as shown in [Barratt & Milnor 1962]. But $\check{H}^3(X;\mathbb{Z}) = 0$ and $\check{H}^2(X;\mathbb{Z}) = \mathbb{Z}^\infty$, the direct sum of countably many copies of $\mathbb{Z}$.

Oddly enough, the corresponding Čech homology groups defined using inverse limits are not so well-behaved. This is because the exactness axiom fails due to the algebraic fact that an inverse limit of exact sequences need not be exact, as a direct limit would be; see §3.F. However, there is a way around this problem using a more refined definition. This is Steenrod homology theory, which the reader can find out about in [Milnor 1995].

Exercises

1. Show that there exist nonorientable 1-dimensional manifolds if the Hausdorff condition is dropped from the definition of a manifold.

2. Show that deleting a point from a manifold of dimension greater than 1 does not affect orientability of the manifold.

3. Show that every covering space of an orientable manifold is an orientable manifold.

4. Given a covering space action of a group G on an orientable manifold M by orientation-preserving homeomorphisms, show that M/G is also orientable.

5. Show that $M \times N$ is orientable iff M and N are both orientable.

6. Given two disjoint connected n-manifolds M_1 and M_2, a connected n-manifold $M_1 \sharp M_2$, their *connected sum*, can be constructed by deleting the interiors of closed n-balls $B_1 \subset M_1$ and $B_2 \subset M_2$ and identifying the resulting boundary spheres ∂B_1 and ∂B_2 via some homeomorphism between them. (Assume that each B_i embeds nicely in a larger ball in M_i.)

(a) Show that if M_1 and M_2 are closed then there are isomorphisms $H_i(M_1 \sharp M_2;\mathbb{Z}) \approx H_i(M_1;\mathbb{Z}) \oplus H_i(M_2;\mathbb{Z})$ for $0 < i < n$, with one exception: If both M_1 and M_2 are nonorientable, then $H_{n-1}(M_1 \sharp M_2;\mathbb{Z})$ is obtained from $H_{n-1}(M_1;\mathbb{Z}) \oplus H_{n-1}(M_2;\mathbb{Z})$ by replacing one of the two $\mathbb{Z}_2$ summands by a $\mathbb{Z}$ summand. [Euler characteristics may help in the exceptional case.]
(b) Show that $\chi(M_1 \sharp M_2) = \chi(M_1) + \chi(M_2) - \chi(S^n)$ if M_1 and M_2 are closed.

7. For a map $f: M \to N$ between connected closed orientable n-manifolds with fundamental classes $[M]$ and $[N]$, the *degree* of f is defined to be the integer d such that $f_*([M]) = d[N]$, so the sign of the degree depends on the choice of fundamental classes. Show that for any connected closed orientable n-manifold M there is a degree 1 map $M \to S^n$.

8. For a map $f: M \to N$ between connected closed orientable n-manifolds, suppose there is a ball $B \subset N$ such that $f^{-1}(B)$ is the disjoint union of balls B_i each mapped homeomorphically by f onto B. Show the degree of f is $\sum_i \varepsilon_i$ where ε_i is $+1$ or -1 according to whether $f: B_i \to B$ preserves or reverses local orientations induced from given fundamental classes $[M]$ and $[N]$.

9. Show that a p-sheeted covering space projection $M \to N$ has degree $\pm p$, when M and N are connected closed orientable manifolds.

10. Show that for a degree 1 map $f: M \to N$ of connected closed orientable manifolds, the induced map $f_*: \pi_1 M \to \pi_1 N$ is surjective, hence also $f_*: H_1(M) \to H_1(N)$. [Lift f to the covering space $\widetilde{N} \to N$ corresponding to the subgroup $\operatorname{Im} f_* \subset \pi_1 N$, then consider the two cases that this covering is finite-sheeted or infinite-sheeted.]

11. If M_g denotes the closed orientable surface of genus g, show that degree 1 maps $M_g \to M_h$ exist iff $g \geq h$.

12. As an algebraic application of the preceding problem, show that in a free group F with basis $x_1, \cdots, x_{2k}$, the product of commutators $[x_1, x_2] \cdots [x_{2k-1}, x_{2k}]$ is not equal to a product of fewer than k commutators $[v_i, w_i]$ of elements $v_i, w_i \in F$. [Recall that the 2-cell of M_k is attached by the product $[x_1, x_2] \cdots [x_{2k-1}, x_{2k}]$. From a relation $[x_1, x_2] \cdots [x_{2k-1}, x_{2k}] = [v_1, w_1] \cdots [v_j, w_j]$ in F, construct a degree 1 map $M_j \to M_k$.]

13. Let $M_h' \subset M_g$ be a compact subsurface of genus h with one boundary circle, so M_h' is homeomorphic to M_h with an open disk removed. Show there is no retraction $M_g \to M_h'$ if $h > g/2$. [Apply the previous problem, using the fact that $M_g - M_h'$ has genus $g - h$.]

14. Let X be the shrinking wedge of circles in Example 1.25, the subspace of $\mathbb{R}^2$ consisting of the circles of radius $1/n$ and center $(1/n, 0)$ for $n = 1, 2, \cdots$.
(a) If $f_n: I \to X$ is the loop based at the origin winding once around the n^{th} circle, show that the infinite product of commutators $[f_1, f_2][f_3, f_4] \cdots$ defines a loop in X that is nontrivial in $H_1(X)$. [Use Exercise 12.]

(b) If we view X as the wedge sum of the subspaces A and B consisting of the odd-numbered and even-numbered circles, respectively, use the same loop to show that the map $H_1(X) \to H_1(A) \oplus H_1(B)$ induced by the retractions of X onto A and B is not an isomorphism.

15. For an n-manifold M and a compact subspace $A \subset M$, show that $H_n(M, M-A; R)$ is isomorphic to the group $\Gamma_R(A)$ of sections of the covering space $M_R \to M$ over A, that is, maps $A \to M_R$ whose composition with $M_R \to M$ is the identity.

16. Show that $(\alpha \frown \varphi) \frown \psi = \alpha \frown (\varphi \smile \psi)$ for all $\alpha \in C_k(X;R)$, $\varphi \in C^\ell(X;R)$, and $\psi \in C^m(X;R)$. Deduce that cap product makes $H_*(X;R)$ a right $H^*(X;R)$-module.

17. Show that a direct limit of exact sequences is exact. More generally, show that homology commutes with direct limits: If $\{C_\alpha, f_{\alpha\beta}\}$ is a directed system of chain complexes, with the maps $f_{\alpha\beta} : C_\alpha \to C_\beta$ chain maps, then $H_n(\varinjlim C_\alpha) = \varinjlim H_n(C_\alpha)$.

18. Show that a direct limit $\varinjlim G_\alpha$ of torsionfree abelian groups G_α is torsionfree. More generally, show that any finitely generated subgroup of $\varinjlim G_\alpha$ is realized as a subgroup of some G_α.

19. Show that a direct limit of countable abelian groups over a countable indexing set is countable. Apply this to show that if X is an open set in $\mathbb{R}^n$ then $H_i(X;\mathbb{Z})$ is countable for all i.

20. Show that $H_c^0(X;G) = 0$ if X is path-connected and noncompact.

21. For a space X, let X^+ be the one-point compactification. If the added point, denoted ∞, has a neighborhood in X^+ that is a cone with ∞ the cone point, show that the evident map $H_c^n(X;G) \to H^n(X^+, \infty; G)$ is an isomorphism for all n. [Question: Does this result hold when $X = \mathbb{Z} \times \mathbb{R}$?]

22. Show that $H_c^n(X \times \mathbb{R}; G) \approx H_c^{n-1}(X;G)$ for all n.

23. Show that for a locally compact Δ-complex X the simplicial and singular cohomology groups $H_c^i(X;G)$ are isomorphic. This can be done by showing that $\Delta_c^i(X;G)$ is the union of its subgroups $\Delta^i(X,A;G)$ as A ranges over subcomplexes of X that contain all but finitely many simplices, and likewise $C_c^i(X;G)$ is the union of its subgroups $C^i(X,A;G)$ for the same family of subcomplexes A.

24. Let M be a closed connected 3-manifold, and write $H_1(M;\mathbb{Z})$ as $\mathbb{Z}^r \oplus F$, the direct sum of a free abelian group of rank r and a finite group F. Show that $H_2(M;\mathbb{Z})$ is $\mathbb{Z}^r$ if M is orientable and $\mathbb{Z}^{r-1} \oplus \mathbb{Z}_2$ if M is nonorientable. In particular, $r \geq 1$ when M is nonorientable. Using Exercise 6, construct examples showing there are no other restrictions on the homology groups of closed 3-manifolds. [In the nonorientable case consider the manifold N obtained from $S^2 \times I$ by identifying $S^2 \times \{0\}$ with $S^2 \times \{1\}$ via a reflection of S^2.]

25. Show that if a closed orientable manifold M of dimension $2k$ has $H_{k-1}(M;\mathbb{Z})$ torsionfree, then $H_k(M;\mathbb{Z})$ is also torsionfree.

26. Compute the cup product structure in $H^*(S^2\times S^8 \sharp S^4\times S^6;\mathbb{Z})$, and in particular show that the only nontrivial cup products are those dictated by Poincaré duality. [See Exercise 6. The result has an evident generalization to connected sums of $S^i\times S^{n-i}$'s for fixed n and varying i.]

27. Show that after a suitable change of basis, a skew-symmetric nonsingular bilinear form over $\mathbb{Z}$ can be represented by a matrix consisting of 2×2 blocks $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ along the diagonal and zeros elsewhere. [For the matrix of a bilinear form, the following operation can be realized by a change of basis: Add an integer multiple of the i^{th} row to the j^{th} row and add the same integer multiple of the i^{th} column to the j^{th} column. Use this to fix up each column in turn. Note that a skew-symmetric matrix must have zeros on the diagonal.]

28. Show that a nonsingular symmetric or skew-symmetric bilinear pairing over a field F, of the form $F^n\times F^n\to F$, cannot be identically zero when restricted to all pairs of vectors v,w in a k-dimensional subspace $V\subset F^n$ if $k>n/2$.

29. Use the preceding problem to show that if the closed orientable surface M_g of genus g retracts onto a graph $X\subset M_g$, then $H_1(X)$ has rank at most g. Deduce an alternative proof of Exercise 13 from this, and construct a retraction of M_g onto a wedge sum of k circles for each $k\le g$.

30. Show that the boundary of an R-orientable manifold is also R-orientable.

31. Show that if M is a compact R-orientable n-manifold, then the boundary map $H_n(M,\partial M;R)\to H_{n-1}(\partial M;R)$ sends a fundamental class for $(M,\partial M)$ to a fundamental class for ∂M.

32. Show that a compact manifold does not retract onto its boundary.

33. Show that if M is a compact contractible n-manifold then ∂M is a homology $(n-1)$-sphere, that is, $H_i(\partial M;\mathbb{Z})\approx H_i(S^{n-1};\mathbb{Z})$ for all i.

34. For a compact manifold M verify that the following diagram relating Poincaré duality for M and ∂M is commutative, up to sign at least:

$$\begin{tikzcd}
H^{k-1}(\partial M;R) \arrow[r] \arrow[d, "{[\partial M]\frown}"] & H^k(M,\partial M;R) \arrow[r] \arrow[d, "{[M]\frown}"] & H^k(M;R) \arrow[r] \arrow[d, "{[M]\frown}"] & H^k(\partial M;R) \arrow[d, "{[\partial M]\frown}"] \\
H_{n-k}(\partial M;R) \arrow[r] & H_{n-k}(M;R) \arrow[r] & H_{n-k}(M,\partial M;R) \arrow[r] & H_{n-k-1}(\partial M;R)
\end{tikzcd}$$

35. If M is a noncompact R-orientable n-manifold with boundary ∂M having a collar neighborhood in M, show that there are Poincaré duality isomorphisms $H_c^k(M;R)\approx H_{n-k}(M,\partial M;R)$ for all k, using the five-lemma and the following diagram:

$$\begin{tikzcd}
\cdots \arrow[r] & H_c^{k-1}(\partial M;R) \arrow[r] \arrow[d, "D_{\partial M}"] & H_c^k(M,\partial M;R) \arrow[r] \arrow[d, "D_M"] & H_c^k(M;R) \arrow[r] \arrow[d, "D_M"] & H_c^k(\partial M;R) \arrow[r] \arrow[d, "D_{\partial M}"] & \cdots \\
\cdots \arrow[r] & H_{n-k}(\partial M;R) \arrow[r] & H_{n-k}(M;R) \arrow[r] & H_{n-k}(M,\partial M;R) \arrow[r] & H_{n-k-1}(\partial M;R) \arrow[r] & \cdots
\end{tikzcd}$$

Additional Topics

3.A Universal Coefficients for Homology

The main goal in this section is an algebraic formula for computing homology with arbitrary coefficients in terms of homology with $\mathbb{Z}$ coefficients. The theory parallels rather closely the universal coefficient theorem for cohomology in §3.1.

The first step is to formulate the definition of homology with coefficients in terms of tensor products. The chain group $C_n(X;G)$ as defined in §2.2 consists of the finite formal sums $\sum_i g_i\sigma_i$ with $g_i \in G$ and $\sigma_i : \Delta^n \to X$. This means that $C_n(X;G)$ is a direct sum of copies of G, with one copy for each singular n-simplex in X. More generally, the relative chain group $C_n(X,A;G) = C_n(X;G)/C_n(A;G)$ is also a direct sum of copies of G, one for each singular n-simplex in X not contained in A. From the basic properties of tensor products listed in the discussion of the Künneth formula in §3.2 it follows that $C_n(X,A;G)$ is naturally isomorphic to $C_n(X,A)\otimes G$, via the correspondence $\sum_i g_i\sigma_i \mapsto \sum_i \sigma_i \otimes g_i$. Under this isomorphism the boundary map $C_n(X,A;G) \to C_{n-1}(X,A;G)$ becomes the map $\partial \otimes \mathbb{1} : C_n(X,A)\otimes G \to C_{n-1}(X,A)\otimes G$ where $\partial : C_n(X,A) \to C_{n-1}(X,A)$ is the usual boundary map for $\mathbb{Z}$ coefficients. Thus we have the following algebraic problem:

> Given a chain complex $\cdots \longrightarrow C_n \xrightarrow{\partial_n} C_{n-1} \longrightarrow \cdots$ of free abelian groups C_n, is it possible to compute the homology groups $H_n(C;G)$ of the associated chain complex $\cdots \longrightarrow C_n\otimes G \xrightarrow{\partial_n \otimes \mathbb{1}} C_{n-1}\otimes G \longrightarrow \cdots$ just in terms of G and the homology groups $H_n(C)$ of the original complex?

To approach this problem, the idea will be to compare the chain complex C with two simpler subcomplexes, the subcomplexes consisting of the cycles and the boundaries in C, and see what happens upon tensoring all three complexes with G.

Let $Z_n = \operatorname{Ker}\partial_n \subset C_n$ and $B_n = \operatorname{Im}\partial_{n+1} \subset C_n$. The restrictions of ∂_n to these two subgroups are zero, so they can be regarded as subcomplexes Z and B of C with trivial boundary maps. Thus we have a short exact sequence of chain complexes consisting of the commutative diagrams

(i)
$$\begin{tikzcd}
0 \arrow[r] & Z_n \arrow[r] \arrow[d, "\partial_n"] & C_n \arrow[r, "\partial_n"] \arrow[d, "\partial_n"] & B_{n-1} \arrow[r] \arrow[d, "\partial_{n-1}"] & 0 \\
0 \arrow[r] & Z_{n-1} \arrow[r] & C_{n-1} \arrow[r, "\partial_{n-1}"] & B_{n-2} \arrow[r] & 0
\end{tikzcd}$$

The rows in this diagram split since each B_n is free, being a subgroup of the free group C_n. Thus $C_n \approx Z_n \oplus B_{n-1}$, but the chain complex C is not the direct sum of the chain complexes Z and B since the latter have trivial boundary maps but the boundary maps in C may be nontrivial. Now tensor with G to get a commutative diagram